M1 Proposal: Smart Hospital and Blood Donor Finder

Integrated Mega Project, Fall 2026 | AI 306, CS 312, CO 208 | Badya University

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1. Problem statement

In an emergency or urgent medical need, people in Cairo usually go to the nearest hospital, even when it lacks the right specialty, has a long wait, or has no stock of the needed blood type. Families then spend hours searching for donors on social media. Our system recommends the most suitable hospital by total time to treatment (travel time plus expected wait, with a specialty match). When a hospital reports a blood shortage, it ranks the eligible, compatible donors who can reach it fastest. It is a decision-support tool, not a replacement for emergency services (ambulance: 123).

2. Objectives

1. Reduce mean time to treatment by at least 15% compared with the nearest-hospital baseline over at least 1,000 simulated scenarios.
2. Recommend only hospitals offering the requested specialty in 100% of test scenarios.
3. For simulated blood requests, find a compatible and eligible donor reachable within 45 minutes in at least 85% of cases, beating a nearest-donor baseline.
4. Deliver a working web application with at least 4 pages, a results dashboard and an AI demo page usable by a non-technical person.

3. Data source, size and licence

Dataset	Source	Size (records × variables)	Licence
Road network and hospitals	OpenStreetMap via `osmnx` (Cairo)	Thousands of edges; 50–150 hospitals	ODbL, cited
Hospital wait logs	**Simulated**, model documented	≈15,000 × 9 (hospital, day, hour, specialty, severity, patients present, wait time, peak flag, blood-stock level)	Synthetic
Blood donors	**Simulated**, model documented	≈2,000 × 8 (ID, blood type, age, weight, area, last donation, availability, response probability)	Synthetic, no real people
Blood requests	**Simulated**	≈1,500 × 6	Synthetic

Simulation model: wait times follow a gamma or lognormal distribution with effects for hour, specialty and severity; blood types follow published ABO/Rh population frequencies (to be cited). A seeded Python script generates the data, and a short document justifies every parameter. No real patient or donor data is collected.

4. Statistical questions

1. Does mean time to treatment differ between our recommendation and the nearest-hospital baseline? (95% CI for the paired difference; paired t-test, Wilcoxon signed-rank as non-parametric check.)
2. Does mean wait time differ between peak and off-peak hours and between specialties? (Welch t-test; Kruskal–Wallis.)
3. Does the proportion of blood requests fulfilled within 45 minutes differ between our donor ranking and the baseline? (CI for proportions; chi-square test.)
4. How do hour of day and patients present affect wait time? (Multiple regression; distribution fit with Q–Q plot.)

5. AI component

Technique: informed search (A*), implemented by hand with a priority queue, plus rule-based reasoning for blood compatibility and donor eligibility.

- **Graph:** Cairo road network. Nodes are intersections; edges carry travel time (length / speed × an hour-of-day congestion factor).
- **Hospital search:** start at the user's location. Each hospital links to one virtual goal node whose edge cost is the expected wait time, plus exclusion or a large penalty if the specialty (or required blood stock) does not match. Heuristic: straight-line distance / maximum speed (admissible).
- **Donor search:** when a hospital needs a blood type, A* ranks donors by travel time to it, after rules filter them: ABO/Rh compatibility, age 18–60, weight $\geq$ 50 kg, at least 56 days since last donation.

- **Input / output:** location, specialty, hour and blood type in; recommended hospital, route, expected time and ranked donors out.
- **Baselines:** nearest hospital by distance; nearest compatible donor by straight-line distance (random notification as a second baseline).
- **Evaluation:** 1,000+ random scenarios with a held-out set, metrics (mean time to treatment, 45-minute fulfilment rate), and error analysis where A* loses to the baseline.

6. System sketches

Web pages (Flask or FastAPI): (1) Home with emergency warning (call 123); (2) Dashboard: hospital map, wait-time distributions, A* vs. baseline charts, blood-stock overview; (3) Find a Hospital: choose location, specialty and time, see the route vs. the nearest hospital; (4) Blood Donation: hospital posts a request, ranked eligible donors, simulated donor registration; (5) Methodology and data (optional).

Database entities (3NF): `hospitals`, `specialties`, `hospital_specialties`, `road_nodes`, `road_edges`, `wait_logs`, `donors`, `blood_types`, `blood_requests`, `recommendation_requests`.

7. Risks and plan B

Risk	Plan B
OSM graph too large or slow	Restrict to a few Cairo districts; cache the graph to disk
Simulated results look circular	Separate generator from algorithm; run sensitivity checks; state as a limitation
Blood donation takes too long	Keep it as rule filter plus ranking on the same A*; drop the registration page
Medical misuse	Prominent disclaimers; no real data; discuss safety, fairness and privacy in Responsible AI
A* gains little over baseline	Report honestly, analyse why, vary scenario difficulty (e.g. peak hours)